

Lab Report – EE-170L Computer Programming (Lab 06)

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SEMESTER: 2nd

LAB: 6 (Programming)

DATE: __20 APRIL 2026 _____

1. Objective

The main objective of this lab is to understand and practice the use of the **for loop** and **nested for loop** in C++. Loops are essential in programming because they help repeat tasks efficiently without writing the same statements multiple times.

In this lab, we focus on:

- The **for loop**
- The **nested for loop**
- Using loops for calculations and pattern printing

By the end of this lab, students should be able to:

- Write programs using the for loop.
- Understand how loop initialization, condition, and update work together.

- Use nested loops to generate tables and patterns.
 - Apply loops to solve problems such as summation and multiplication tables.
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2. Theory

2.1 For Loop

The **for loop** is a repetition control structure that allows a block of code to execute multiple times. It is mostly used when the number of iterations is known in advance.

General Syntax:

```
for(initialization; condition; update)
{
    statements;
}
```

Key Points:

- **Initialization** sets the starting value of the loop variable.
- **Condition** determines whether the loop continues or stops.
- **Update** changes the loop variable after each iteration.

The for loop is compact because all three parts are written in one line.

2.2 Nested For Loop

A **nested for loop** means placing one for loop inside another. The outer loop controls the number of rows, while the inner loop controls the columns or repeated operations inside each row.

Nested loops are commonly used for:

- Printing patterns
 - Multiplication tables
 - Working with rows and columns
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3. Programs and Outputs

Task 1: Print numbers from 1 to 10 using for loop

```
#include <iostream>
using namespace std;

int main()
{
    for(int i = 1; i <= 10; i++)
    {
        cout << i << endl;
    }

    return 0;
}
```

Output:

```
1
2
3
4
5
6
7
8
9
10
```

Task 2: Multiplication Table using Nested for Loop

```
#include <iostream>
using namespace std;

int main()
{
    for(int i = 1; i <= 10; i++)
    {
        for(int j = 1; j <= 10; j++)
        {
```

```

        cout << i * j << "\t";
    }

    cout << endl;
}

return 0;
}

```

Output:

```

1 2 3 4 5 ...
2 4 6 8 10 ...
3 6 9 12 15 ...

```

(Multiplication table from 1 to 10 displayed.)

Task 3: Sum of numbers from 1 to n using for loop

```

#include <iostream>
using namespace std;

int main()
{
    int n, sum = 0;

    cout << "Enter a positive integer: ";
    cin >> n;

    for(int i = 1; i <= n; i++)
    {
        sum += i;
    }

    cout << "Sum = " << sum << endl;

    return 0;
}

```

Output:

Enter a positive integer: 5
Sum = 15

Task 4: Triangle Pattern using Nested for Loop

```
#include <iostream>
using namespace std;

int main()
{
    int rows;

    cout << "Enter number of rows: ";
    cin >> rows;

    for(int i = 1; i <= rows; i++)
    {
        for(int j = 1; j <= i; j++)
        {
            cout << "* ";
        }

        cout << endl;
    }

    return 0;
}
```

Output:

```
*
* *
* * *
* * * *
* * * * *
```

4. Discussion

This lab demonstrated the importance of loops in programming. The **for loop** is very useful when the number of repetitions is already known. It makes the code shorter, cleaner, and easier to understand.

- The **for loop** combines initialization, condition, and update in one statement, making it compact and efficient.
- The **nested for loop** is powerful for generating tables and patterns because one loop can control rows while the other controls columns.
- Using loops helps reduce code repetition and improves programming efficiency.

By completing these tasks, I learned how loops control program execution and how nested loops can solve more complex problems like multiplication tables and pattern printing.

5. Conclusion

In this lab, I practiced using the **for loop** and **nested for loop** in C++. I learned how loops can repeat tasks automatically and reduce unnecessary code repetition.

The for loop is useful when the number of iterations is known, while nested loops help in creating tables and patterns.

This lab improved my understanding of loop structures and strengthened my programming skills in C++.